

KSHETRA — Research Compendium

Automated Integration and Intelligent Harmonization of Multi-Source Geospatial Data for Urban Land Record Management

Team NovaX · Smart India Hackathon 2026 · Problem Statement 26013 · 22 commits · 13,697 lines of first-party code

Three documents exist for this project. The *Research Summary* (9 pp) is the literature-facing extract. The *Project Dossier* (22 pp) is the engineering record. This is the complete one: the problem in its historical and legal depth, the literature read closely, **the method written out as mathematics rather than described**, a full feature dictionary, the evaluation in detail, a formal comparison against prior art, and a glossary of the domain vocabulary a reviewer outside Indian land administration will need.

Conventions

Claims about the world are graded. [V] verified against a primary or peer-reviewed source. [S] secondary — consistently reported, not traced to the original. [U] unverified. Appendix G lists the circulating statistics we decline to use and why.

Claims about our own system are measured. Every figure in Parts D and E is read from `data/demo/*.json` when this PDF is built; line counts come from `git ls-files` and the commit table from `git log`. The document cannot disagree with the repository.

Document map

Part	Covers	Read it for
A	The problem domain — history, law, vocabulary, programme, scale	Why this is hard for reasons that are not technical
B	Literature and tooling	What is solved, what is not, and who established it
C	Method — the mathematics of all thirteen stages	Exactly what the system computes
D	Evaluation	How we know, including what we got wrong first
E	System, deployment, engineering record	What was actually built and what it cost
F	Assessment — comparison, limits, validity, roadmap	Where this stands and where it does not

Abstract

NAKSHA specifies new urban orthoimagery to **RMSE ≤ 10 cm**. The legacy cadastral sheet it must be reconciled with has a modal positional error of **3–4 m** [5]. Urban plots are of the order of **11 m** across, so an unaligned legacy parcel overlaps its *neighbour* more than its own counterpart: naive overlay is not imprecise, it is meaningless. We frame the problem as **spatial entity resolution under a 30–40× accuracy mismatch** rather than as boundary extraction, and address three gaps the literature leaves open. First, **no ground truth exists**, so we manufacture it — generating a clean cadastre and damaging it through twelve stages drawn from a documented failure taxonomy, following the known-deformation principle of [2]. Second, **confidence is usually a weighted score**, so we calibrate ours by isotonic regression and report expected calibration error of **0.00009**, which is what makes an explicit triage threshold possible. Third, **ground truthing is NAKSHA's most delayed component** [9], so we close the loop: a Gaussian process over the residual error field plus greedy submodular maximisation with the $(1 - 1/e)$ bound of [8] selects where to survey next, and the recommendation is *validated* by applying it rather than asserted. On a held-out synthetic city the matcher takes **95.6%** of its decision from geometry at F1 **0.9993**, topology repair eliminates all **5,038** overlapping parcel pairs while an area ledger demonstrates no land was created or destroyed, and the planner achieves **1.3177 m** against **2.0567 m** for random placement of the same budget.

Part A — The problem domain

Why Indian cadastral harmonisation is hard for reasons that predate computing.

A1. Historical genesis of the fragmentation

The Survey of India was established in **1767** and conducted revenue survey nationally until **1904**, after which **each state became responsible for its own cadastral survey and evolved its own system** [10] [V]. Every interoperability problem in this problem statement descends from that single administrative devolution: divergent vocabulary, divergent units, divergent sheet scales, divergent legal primacy, divergent numbering grammar. There is no technical fix for a 120-year institutional divergence; there is only a system designed to expect it.

A2. Legal primacy differs between north and south

Northern states recognise the **graphic map** as the legal document. Southern states prioritise the **Field Measurement Book** [11] [V].

The consequence is not academic. In the north, the polygon carries legal authority, so correcting its geometry corrects the record. In the south, the *measurements* carry authority and the polygon is a derived sketch, so rewriting the polygon changes a drawing, not a right — and a system that treats its output as authoritative is destroying the record in one half of the country while merely correcting it in the other.

This is the direct reason our conflict resolver separates *measured* fields from *recorded* fields and applies a different rule to each (§C12), and why a per-state policy object appears in the roadmap rather than a global configuration flag.

A3. Semantic heterogeneity is structural, not lexical

A synonym table is insufficient, for three separate reasons.

Numbering carries grammar

South Indian survey numbers encode a recursive sub-division tree ($123 \rightarrow 123/1 \rightarrow 123/1A$). A matcher that treats 123 and $123/1A$ as a mismatch is discarding the single clearest signal that a subdivision occurred. We encode this as `kh_parent` and `kh_child` features (§C6).

The same concept has different names

The record of rights is the *khatauni* in UP, the *jamabandi* in Punjab and Haryana, the *patta/chitta* in Tamil Nadu. The holder is the *khatedar*, *pattadar* or *bhogvatadar*. The parcel number is *khasra*, *survey number*, *gat number* or *kitta*.

The same name has different concepts — the dangerous case

"Khata" denotes a revenue holding account in the north but a municipal property-tax account in urban Karnataka. A lexicon that maps the token to one canonical field will silently produce wrong joins in an urban project. This is why our schema matcher profiles *values* as well as reading names (§C11): a column with four distinct values across 3,000 rows is categorical and cannot be a holding identifier, whatever it is called.

A4. Programme and policy context

Programme	What it is	Bearing on this work
NAKSHA	Urban land-record modernisation; drone survey of pilot ULBs	Produces the 10 cm layer. Its progress review [9] names cross-agency non-interoperability as a systemic void and ground truthing as the most delayed component — the two gaps we aim at.
DILRMP	Digital India Land Records Modernisation Programme	Source of the 49.10% geo-referenced-village figure [13] and of ULPIN.
SVAMITVA	Drone survey of inhabited rural <i>abadi</i> land	Demonstrates the drone-to-record pipeline at national scale; rural, so adjacent rather than identical.
ULPIN	14-character parcel identifier from geo-coordinates	We implement a faithful <i>shape</i> ; the exact composition rule is [U] and we do not invent it.
NGP 2022	National Geospatial Policy [7]	Commits to <i>redefining</i> the national geodetic framework. Note it does not create an "Indian Geodetic Datum 2023"; see Appendix G.

A4.1 What the new data will look like

Survey of India technical circular T-260/1147-Project, 10 February 2025 [6] [V]: orthorectified imagery at **5 cm GSD**, **RMSE(x,y) ≤ 10 cm**; DSM/DTM at 0.5 m spacing, **RMSE(z) ≤ 15 cm**; CORS control better than 5 cm; 70%/60% overlap; minimum four corner and one centre GCP. CRS settled as **UTM on WGS 84**, vertical on the Indian Vertical Datum via the Sol geoid model.

A5. The problem quantified

Several widely-circulated Indian land-dispute statistics are recycled with drifting attribution. Only traceable figures appear below.

Claim	Source	Grade
7,933+ urban settlements, ~10.2 M ha, 4,912 ULBs	[9, 12]	[V]
Only four states maintain structured urban land records (TN, MH, GJ, GA)	[9] and DoLR's NAKSHA booklet [12], independently	[V]
~13 M urban households in 108,000 slums	[9]	[V]
49.10% of villages hold geo-referenced cadastral maps	[13]	[V]
Land market distortions erode ~1.3% of annual GDP growth	[9]	[V]
ULBs generate only 32% of revenue internally	[9]	[V]
Land/property taxes 0.6% of GDP in low-income vs 2.2% industrialised	[9]	[V]
Property disputes 60–70% of Indian civil litigation	[9]	[V] as a peer-reviewed statement
~Two-thirds of surveyed civil matters concerned land and property	DAKSH 2015-16; 9,329 litigants, 305 locations, 24 states [14]	[V] as a survey of litigants, not a census
Resurvey rate ₹56,725 / sq km	Tamil Nadu, DILRMP review 2024 [15]	[V]
Resurvey takes 2–3 years per district tranche even with drone + DGPS	[15]	[V]

A6. How wrong the legacy maps are — the central measurement

Sengupta, Lemmen, Devos, Bandyopadhyay and van der Veen georeferenced **310 analogue cadastral sheets** at 1:3960 over a 327 km² planning area in West Bengal — 258 mouzas, 26 municipal wards, mapped mostly from pre-1920s surveys and 1950s work — against GeoEye-1 pan-sharpened imagery with 10–15 GCPs per sheet and a first-order transformation [5] [V].

RMSE band	Sheets	Share	Cumulative
< 2.00 m	7	2.3%	2.3%
2.01 – 3.00 m	64	20.6%	22.9%
3.01 – 4.00 m	185	59.7%	82.6%
4.01 – 5.00 m	47	15.2%	97.7%
> 5.01 m	7	2.3%	100%

The **modal** error is 3–4 m and **88% of sheets fall between 2 and 5 m**. This is the single most load-bearing number in the project: it is what makes the problem an entity-resolution problem rather than an overlay problem.

A6.1 Transformation order is still chosen by a human

Sample	1st order	2nd order	Reduction
1	3.668 m	3.177 m	13%
2	3.171 m	2.510 m	21%
3	3.962 m	1.339 m	66%
4	4.089 m	3.096 m	24%
5	5.010 m	2.889 m	42%

Up to two thirds of the error is removable by a higher-order transformation — but choosing the order per sheet is manual [5]. §C3 describes our escalating selector, which is the direct response.

A6.2 Documented failure taxonomy [5] [V]

- **Gaps and overlaps between adjoining mouza sheets** — adjacent sheets do not fit.
- **Multi-sheet mouzas** with unclear division between sheets.
- **Canal double-counting** — a shared boundary drawn on both sheets, so area is counted twice.
- **River dynamics** — land lost or accreted, giving large legal-versus-digitised discrepancies.
- **Paper degradation** — shrinkage, wrinkling, folding, tearing; scanning adds more.
- **Crude map lines** — scanned lines over 4–5 px wide, so boundary ambiguity of decimetres to metres exists *before* any transformation.
- **Area attribute inconsistency** — digitised area $\neq$ authorised area in the RoR.

Our damage harness (§D2) reproduces five of these seven directly. That it derives from a documented taxonomy rather than from imagination is why we consider the recovery figures meaningful.

A7. The mismatch, stated formally

New layer $\sigma \approx 0.10$ m. Legacy layer $\sigma \approx 3.5$ m. Characteristic parcel width $w \approx 11$ m.

For a legacy parcel displaced by d , expected overlap with its true counterpart falls roughly as $(1 - d/w)^2$ while overlap with the adjacent parcel *rises*. The two cross near **$d = w/2 \approx 5.5$ m** — comfortably inside the observed 2–5 m band once warp and digitising noise compound. Beyond that crossover the nearest-overlap heuristic that every overlay tool relies on is not merely noisy, it is **systematically wrong**, and confidently so.

This is the whole argument for the pipeline order in §C1, and we measured it rather than assuming it: matching before georeferencing gave blocking recall **0.58**; after reordering, **1.0** (§D4).

Part B — Literature and tooling

Five papers read in the original; the tool landscape; the gap.

B1. Chen, Nazeer, Lee & Wong 2026 — the systematic review [1]

Published March 2026 in *Land*. Establishes that deep learning has removed the manual-interpretation bottleneck in boundary extraction via CNN and Transformer segmentation, that NLP methods now extract structure from paper archives, and that deep models detect parcel change and support integrated spatial/non-spatial analysis [V]. Its closing statement of open problems:

"...challenges remain, including differences in multi-temporal data processing, spatial semantic ambiguity, and the lack of large-scale, high-quality annotated data. Future research can focus on improving model generalization, advancing cross-modal data fusion, and providing recommendations for the development of a reliable and practical intelligent cadastral system."

Mapped onto this project: *lack of annotated data* → §D1 manufactured ground truth. *Cross-modal fusion* → §C12 provenance-weighted resolution. *Generalization* → §D3 held-out city evaluation. *Spatial semantic ambiguity* → §C11

schema inference.

B2. Girard, Charpiat & Tarabalka 2019 — alignment without truth [2]

The closest published statement of our core problem. A multi-resolution U-Net predicts a dense 2-D displacement field aligning misregistered cadastral polygons to imagery where no correct annotation exists, trained over rounds that re-correct the annotations between rounds. **Alignment error fell by more than 3× by round two; round three added nothing [V].**

The method we borrow

Training data is constructed by applying *known random deformations* and learning to invert them — formally a dataset

$$D = \{ (I, J_{\text{rand}}, f_{\text{rand}}) \}$$

where the deformation f is generated and therefore known. Our damage harness is this principle, applied to an Indian failure taxonomy instead of generic warps. This matters for how we defend it: manufactured ground truth is **published method**, not an improvisation forced on us by data scarcity.

The two limitations we inherit

A perfect alignment score is *unattainable* because the reference annotations are themselves ambiguous — many buildings are outlined by a coarse polygon, so "best alignment" is ill-posed with multiple equally good solutions. And the model learns only **smooth** displacement fields, so adjoining buildings requiring a discontinuous correction fail.

Our Gaussian process carries exactly the same smoothness assumption — a squared-exponential kernel cannot represent a sheet-join discontinuity. We therefore report an irreducible error floor rather than implying arbitrary accuracy is purchasable with survey budget (§C13).

B3. Suwardhi et al. 2025 — the closest operational system [3]

UAV orthophotos at 5 cm GSD segmented by the Segment Anything Model without manual prompting; displacement vectors from ICP; hierarchical adjustment from rigid block (LS1) to scaled block (LS2) to individual parcel (LS3). Evaluated over two urban villages in Cimahi, Indonesia — **Karangmekar**, 81 blocks / 3,227 parcels and **Baros**, 96 blocks / 2,971 parcels — chosen for contrasting block structure and data quality. Client-server, Python REST API [V].

What it cannot do, and why that is the instructive part

Because no correct geometry exists for those areas either, the authors evaluate **by proxy**: they count how often cadastral polygons split the SAM-derived boundary segments, treating fewer splits as better alignment. That measures *internal consistency*, not accuracy. **The state of the art cannot report its error in metres.**

It is also single-source and single-country, and addresses none of: attribute schema inference, CRS/datum/unit heterogeneity, tenure semantics, calibrated confidence, or where to survey next.

B4. Fetai, Grigillo & Lisec 2022 — boundary revision [4]

A modified CNN detects visible land boundaries from image-based mapping and uses them to revise existing cadastral data [V]. This sits *upstream* of our pipeline and we deliberately do not rebuild it: extraction is well served by the literature, and the problem statement asks for integration of extracted features, not another extractor. Building a worse version of a solved problem would be the easiest way to lose this hackathon.

B5. Krause, Singh & Guestrin 2008 — the imported theorem [8]

Sensor placement in Gaussian processes. With S the chosen observation set and $F(S)$ the total posterior-variance reduction over the domain, **F is monotone and submodular**, so greedy selection is within $(1 - 1/e) \approx 63\%$ of optimal, and exact optimisation is NP-hard [V].

The result is standard in machine learning and we claim no part of it. What we claim is the transfer: **we found no application of it to cadastral ground-control planning**, and the fit is unusually clean because posterior variance depends only on *where* you observe, never on what you measure — so the entire survey plan is computable before anyone leaves the office (§C14).

B6. LADM and the question posed in 2013

ISO 19152-1:2024 [16] [V]. Explicitly *not* a data product specification: its stated purpose is "not to replace existing systems, but rather to provide a formal language for describing them." Four core packages: parties; basic administrative units carrying rights, restrictions and responsibilities; spatial units; spatial sources and representations.

Sengupta et al. [17] [V] asked in 2013 how to convert colonial maps and records into an LADM database, **how to document and publish the geometric quality of existing maps**, and **how to integrate more accurate data after re-survey**. That final question — fusing a high-accuracy new survey with a low-accuracy legal record without destroying the legal record — is this problem statement, posed in the literature thirteen years ago and still open.

B7. The tooling landscape

Tool	Genuinely does	Does not do
GRASS <code>v.clean</code>	Break/clean polygons from non-topological formats; remove sub-threshold areas by dissolving into the longest-shared-boundary neighbour; fuzzy snapping; vertex pruning	A human picks the thresholds. Wrong value destroys parcels or leaves slivers. A tool, not a decision-maker.
PostGIS Topology	Persistent topological model with shared edges and faces; prevents gaps and overlaps by construction going forward	Getting dirty legacy geometry <i>into</i> a valid topology is the hard part; common practice is to clean in GRASS first.
QGIS + GDAL/PROJ	Georeferencer with polynomial 1st–3rd order, TPS, projective, Helmert; topology checker; reprojection	Manual, per layer, per operator, GCPs by hand. No cross-source reasoning, no confidence, no triage.
FME / Esri Data Interop.	Industry-standard spatial ETL; hundreds of formats; visual schema mapping	The mapping is declared by a human, not inferred. FME executes a mapping faithfully; it will not tell you <code>KHASRA_NO</code> and <code>SY_NO</code> are one concept, nor that they disagree for 12% of parcels. Proprietary, at 4,912-ULB scale.
Esri Parcel Fabric	Mature cadastral editing and QA	Deed-based COGO assumptions; not built for khasra / FMB / mouza semantics or a 30–40× inter-layer mismatch.
SAM / Mask R-CNN / U- Net	Strong at extracting <i>visible</i> boundaries	Cadastral boundaries are frequently invisible — no wall, no fence, no hedge. The ceiling of pure vision.
Bhu-Naksha (incumbent)	National cadastral map software in use across many states	Hand-entered per-state scale factors on shapefile import — UP ×4000, Himachal ×22 where <i>karam</i> was digitised in centimetres — plus hand-written per-state RoR adapters. The incumbent national answer to unit heterogeneity is a magic number. [V]

B8. Synthesis — the eight open problems

#	Open problem	Our response
1	No automated, evidence-weighted reconciliation across accuracy tiers	§C12 — inverse-variance for measured, legal authority for recorded
2	Transformation-model selection is a human judgement	§C3 — escalation driven by residual structure

3	Schema matching for Indian land attributes is entirely manual	§C11 — lexicon + value profiling + unit inference from geometry
4	Topology cleaning has no legal-consequence awareness	§C9 — parcel-sized holes flagged, not filled; partial
5	Vision finds visible boundaries; cadastres contain invisible ones	Not addressed. Named in §F5 as open
6	Cross-agency entity resolution unaddressed [9]	§C4–C8 — the core of the system
7	State of the art is single-source, single-country	§C12 — n-source, per-attribute authority
8	No open benchmark or ground truth exists	§D1 — manufactured; release named in §F5

Five of eight addressed, one partially, one explicitly not. Stating which is which is more useful to a reviewer than claiming all eight.

Part C — Method

What the system actually computes, stage by stage.

C1. Pipeline, and why the order is not negotiable

ingest → schema inference → validate → georeference → blocking → match → assign → topology → conflict → change → uncertainty → targeting → serve

Georeference precedes match by the crossover argument of §A7 and the measurement of §D4. **Schema inference precedes validation** because an area column cannot be range-checked until its unit is known, and the unit is inferred from geometry (§C11), which requires geometry to exist. **Topology precedes conflict** because an overlap is a geometric fact to be removed, not a claim to be adjudicated. **Uncertainty precedes targeting** trivially, but the pairing is the point: targeting is only meaningful because the preceding stages produce an honest error field rather than a point estimate.

C2. Coordinate reference systems

Implemented from Snyder, *Map Projections: A Working Manual* (USGS PP 1395), covering exactly what Indian urban cadastral work needs: WGS84 ↔ UTM for zones 42N–47N, and Everest 1830 (India) ↔ WGS84 via a seven-parameter Helmert transform.

Why Everest 1830 matters. Legacy sheets and older Survey of India products are referenced to it; drone and GNSS output is WGS84. Ignoring the datum difference leaves a **systematic offset of several hundred metres** — two orders of magnitude larger than the misregistration everything else in this document is about, and the single largest error source when overlaying an old map on new imagery. A pipeline that gets this wrong produces beautifully calibrated nonsense.

Ellipsoid parameters derive from a and $1/f$, with first eccentricity

$$e^2 = f(2 - f), \quad f = 1 / \text{inv_f}$$

Helmert is the standard seven-parameter form (three translations, three rotations, one scale) applied in geocentric Cartesian coordinates.

Verified: round-trip < 1 mm; central meridian easting exactly 500,000 m; agreement with pyproj < 1 mm where pyproj is available. Written from scratch because pyproj was blocked by Application Control (§E5), and *kept* because sub-millimetre agreement with the reference implementation is a stronger claim than having imported it.

C3. Coarse alignment

Three stages, each robust to the failure mode of the one before it.

Stage 1 — translation voting (Hough)

Every source centroid votes for the displacement to each of its k nearest reference centroids. The true translation is

shared by thousands of parcels while wrong pairings scatter, so it appears as a sharp peak in a 2-D histogram of votes.

Chosen over ICP deliberately: a Hough transform has **no initialisation requirement and cannot fall into a local minimum**. ICP, which [3] uses, requires a starting estimate good enough that nearest-neighbour correspondences are mostly right — precisely the assumption that §A7 shows fails at 5–15 m displacement on 11 m parcels.

Stage 2 — robust similarity refinement

From the voted translation, fit rotation, scale and translation on *mutual*-nearest-neighbour pairs, reject outliers by a trimmed residual threshold, iterate.

Stage 3 — residual reporting and transform escalation

What remains is the non-linear part — paper warp — removed later using matched parcels as control points. The transform escalates similarity → affine → second-order polynomial, accepted only when residual structure justifies the added degrees of freedom. This is the automated answer to §A6.1.

Reported at the data centroid, not as raw tx/ty. With UTM northings near 3.1 million a raw translation parameter is unauditable; a displacement at the centroid is a number a revenue officer can hold against a map. Small decision, large difference to whether anyone trusts the output.

Quantity	Injected	Recovered
Displacement	(6.5, -4.2) m	7.37 m
Rotation	0.85°	-0.6163°
Scale	1.0016	1.001137
RMSE	8.4529 m	3.127 m
Inliers	—	2,133

C4. Blocking

Exhaustive pairing of $3,047 \times 2,952$ is 9.0 million pairs; we evaluate 52,468. The ranking function matters more than the count:

$$\begin{aligned} \text{rank}(j) &= (1, \text{area}(A \cap B_v)) \quad \text{if overlap} > 0 \\ \text{rank}(j) &= (0, -\|c_v - c_a\|^2) \quad \text{otherwise} \end{aligned}$$

Candidates with any overlap outrank all candidates with none, and within each tier ordering is by overlap area or by centroid proximity respectively. The second tier is what prevents catastrophic recall loss: a parcel whose counterpart was *deleted*, or displaced beyond overlap, still gets its nearest neighbours considered rather than being silently dropped.

Measured blocking recall: **1.0**.

C5. The feature set — complete dictionary

29 features in seven groups. The context group is the one most systems omit and the one that makes calibration possible.

Feature	Group	Definition and why it is here
iou	Overlap	$ A \cap B / A \cup B $. The classic, and alone it is not enough: a displaced parcel and its neighbour can score similarly.
sym_diff_norm	Overlap	$ A \Delta B / (A + B)$. Penalises disagreement symmetrically; sensitive where IoU saturates.
src_frac_in_tgt	Overlap	$ A \cap B / A $. Asymmetric containment. Highest-gain feature in the trained model because it detects subdivision: a child parcel is fully inside its parent even though IoU is low.

<code>tgt_frac_in_src</code>	Overlap	$ A \cap B / B $. The mirror. Together the pair distinguishes split from merge from 1:1.
<code>centroid_dist</code>	Position	Euclidean distance between centroids, metres. Second-highest gain.
<code>centroid_dist_norm</code>	Position	Centroid distance divided by the square root of mean area — scale-free, so a 3 m error means something different for a 50 m ² plot than a 5,000 m ² one.
<code>hausdorff_norm</code>	Position	Normalised Hausdorff distance. Catches the case where centroids agree but boundaries do not.
<code>sig_dist</code>	Position	Distance between turning-function shape signatures; rotation-invariant.
<code>area_ratio</code>	Shape	$\min(A , B) / \max(A , B)$. Bounded in (0, 1] so it cannot dominate.
<code>d_compactness</code>	Shape	Difference in $4\pi A/P^2$. A digitised parcel and its counterpart share compactness even when displaced.
<code>d_rectangularity</code>	Shape	Difference in area over minimum-rotated-rectangle area. Urban plots are near-rectangular; this is informative here in a way it would not be rurally.
<code>d_elongation</code>	Shape	Difference in minimum-rectangle aspect ratio.
<code>vertex_ratio</code>	Shape	Ratio of vertex counts. Decimation during digitisation changes this, so it is weak evidence, carried for completeness.
<code>name_jw</code>	Owner	Jaro-Winkler on owner names after transliteration folding.
<code>name_lev</code>	Owner	Normalised Levenshtein distance.
<code>name_tok</code>	Owner	Token-set ratio, which survives reordering: <i>Sita Tiwari d/o Ajay</i> against <i>Tiwari Sita</i> .
<code>name_present</code>	Owner	Whether a name exists on both sides at all. A missingness signal, not a join key.
<code>kh_exact</code>	Khasra	Exact string equality of the khasra/survey number.
<code>kh_parent</code>	Khasra	Whether the target number is the parent of the source under the recursive grammar (123/1A $\rightarrow$ 123).
<code>kh_child</code>	Khasra	The converse. Together these encode subdivision as evidence rather than as mismatch.
<code>kh_present</code>	Khasra	Whether a number exists on both sides. Again presence, not value.
<code>area_attr_ratio</code>	Attribute	Ratio of <i>recorded</i> areas after unit normalisation — independent of the geometry, so it is genuine corroboration.
<code>land_use_match</code>	Attribute	Land-use code agreement after mapping to a canonical vocabulary.
<code>ward_match</code>	Attribute	Administrative ward agreement.
<code>n_candidates</code>	Context	How many candidates this source parcel has. The model needs to know whether it is choosing among two or among twenty.
<code>iou_rank</code>	Context	Rank of this pair's IoU among the source's candidates.
<code>iou_margin</code>	Context	Gap between this IoU and the runner-up. The ambiguity signal. IoU 0.6 with a margin of 0.4 is a different situation from IoU 0.6 with a margin of 0.01, and without this the model cannot tell them apart.
<code>is_best_iou</code>	Context	Whether this is the source's top candidate by overlap.
<code>rev_iou_rank</code>	Context	Rank from the target's point of view. Mutual-best is far stronger evidence than one-way-best.

C6. Why khasra features are structural, not identity

Note what the four khasra features encode: `kh_exact` (equality), `kh_parent` / `kh_child` (position in the sub-division tree), and `kh_present` (existence). **None of them is the number itself.** The model cannot memorise a lookup from khasra to parcel; it can only use the number as weak corroboration and as a subdivision signal.

This is deliberate and it is tested. §D4 describes the first model, which scored F1 0.9955 by doing exactly what this design forbids.

C7. Model and calibration

Why gradient-boosted trees

The input is ~29 heterogeneous tabular features with strong monotone structure and heavy interaction — IoU matters far more when candidate count is low; name similarity matters far more when IoU is ambiguous. Boosted trees are the right tool for that shape, they train in seconds, and **the decision path for any individual parcel can be printed and defended**, which a land-administration audit trail requires and a neural network does not provide.

Why calibration is a separate stage

Raw GBDT margins through a sigmoid are systematically overconfident. Isotonic regression on a held-out split maps them onto frequencies that actually hold. Formally, isotonic regression finds the monotone non-decreasing $\hat{g}$ minimising

$$\sum_i w_i (y_i - \hat{g}(s_i))^2 \quad \text{subject to} \quad s_i \leq s_j \Rightarrow \hat{g}(s_i) \leq \hat{g}(s_j)$$

Monotone because a higher score should never mean a lower probability; non-parametric because we have no reason to believe the miscalibration is sigmoidal.

Group splitting

Splits are made **by source parcel, never by pair**, so one parcel's candidates cannot straddle train and test. Splitting by pair would leak: the model would see parcel P's correct match in training and its near-misses in test, and every metric would be inflated.

C8. Calibration metrics, defined

Hand-implemented because scikit-learn was blocked (§E5), which forced us to state precisely what we were computing.

$$\text{ECE} = \sum^n (n^b/N) \parallel \text{acc}(b) - \text{conf}(b) \parallel$$

over equal-width bins b . Measured: **0.00009**.

$$\text{Brier} = (1/N) \sum_i (p_i - y_i)^2$$

Measured: **0.00008**. Brier decomposes into reliability minus resolution plus uncertainty, so a low Brier with a low ECE means the model is both calibrated *and* discriminative — a constant predictor achieves perfect calibration and useless resolution, which is why neither number is reported alone.

Calibration gets as much attention in this codebase as accuracy, deliberately. The purpose of the system is **triage**: an officer finalises the confident matches and reviews the rest. A model that is 95% accurate but claims 99% confidence everywhere is useless for triage, because no safe threshold exists.

C9. Assignment

Pairwise scores are not enough. A cadastre is a **partition of space**, so the mapping must be globally coherent: one parcel cannot match three others because all three scored well individually.

Containment is tested **before** the Hungarian assignment, because a subdivided parcel is not a failed 1:1 match and forcing it through a 1:1 solver destroys exactly the information a cadastral officer most needs.

Relation	Meaning	Count
one_to_one	Unchanged between layers	2,791
split	Subdivision — a holding sold off in parts	148
merge	Amalgamation — holdings combined	5
unmatched_source	Genuinely new, or spurious	103
unmatched_reference	Present in reference, missing from source	77

Thresholds (containment 0.50, split/merge coverage 0.55) were chosen for **precision over F1**: a wrongly asserted

subdivision is a worse outcome in a land record than an unresolved case sent to a human.

C10. Topology as a planar partition

A cadastre tiles the ground with no gaps and no overlaps. Layers digitised parcel-by-parcel from paper never satisfy this, and the violations are not cosmetic: **an overlap means two people are recorded as owning the same square metre, and a gap means land that legally belongs to nobody.**

Five stages, fixed order because each assumes the previous succeeded:

1. **Validity** — self-intersecting rings repaired first; area on a bow-tie is meaningless, so every later stage depends on this.
2. **Vertex hygiene** — duplicate and near-collinear vertices removed.
3. **Vertex snapping** — the workhorse. Vertices from different parcels within tolerance are clustered by union-find over cKDTree pairs and collapsed onto one point. This closes hairline gaps and removes hairline overlaps *simultaneously*, because both arise from one cause: two surveyors digitising one shared boundary twice.
4. **Residual overlap resolution** — contested area awarded by an explicit auditable rule, not by whichever polygon is drawn last.
5. **Gap classification** — see below.

19,277 vertices snapped on the evaluation city.

C10.1 The refusal

The stage that matters most is the one that declines to act. Residual gaps are classified by area, compactness and whether anything in the record claims them. Sliver artefacts are closed; **parcel-sized holes are flagged for survey and left open**, because filling one fabricates a boundary and awards land to whoever happens to adjoin it.

Of 73 parcels the harness deleted to create holes, the cleaner **refused 69**. A verification assertion enforces this, because a cleaner that fills everything reports fewer errors and would otherwise look like an improvement (§D5).

C11. Schema inference

Three independent signals, because none alone is reliable.

Name similarity against an Indian revenue lexicon. Strong when the column is conventionally named; useless when it is F14 or COL_3.

Value profiling — dtype, cardinality, uniqueness, regex shape. A column 99% unique matching `\d+ (/ \d+)?` is a khasra number whatever it is called; a column with four distinct values across 3,000 rows is categorical and cannot be an owner name.

Unit inference against geometry — the part that matters most. Recorded area is meaningless until the unit is known, and records are kept in m², bigha, biswa, acres or square yards. Rather than trust a name, divide recorded values by the *surveyed* area of the same parcels and take the modal ratio:

$$\hat{u} = \text{mode}_i \left(\text{area_recorded}_i / \text{area_geometric}_i \right)^{-1}$$

Measured median **2500.45 m²/unit** against bigha at 2529.285 → inferred **bigha** at confidence **0.9672**, with no human declaring it.

Not academic. NIC's own Bhu-Naksha manual documents a manually entered per-state scale factor on shapefile import — UP ×4000, Himachal ×22. Inferring it removes a hand-entered magic number from the critical path of the incumbent national system.

Assignment is **1:1 by the Hungarian algorithm**, so two columns cannot both claim to be the owner field. Result: **7/7**.

Canonical	Column	Conf.	Evidence
-----------	--------	-------	----------

verification assertion checks is physically plausible (5–500 m).

Posterior variance and mean take the standard forms, solved by Cholesky:

$$\sigma^2(\mathbf{x} | S) = \mathbf{k}(\mathbf{x}, \mathbf{x}) - \mathbf{k}_{\mathbf{x}\mathbf{s}}^T (\mathbf{K}_{\mathbf{ss}} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{k}_{\mathbf{x}\mathbf{s}}$$

$$\mu(\mathbf{x} | S) = \mathbf{k}_{\mathbf{x}\mathbf{s}}^T (\mathbf{K}_{\mathbf{ss}} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{r}_{\mathbf{s}}$$

The key property: posterior variance depends only on *where* you observe, never on what you measure. Variance reduction is therefore computable before any surveyor leaves the office.

Three corrections were needed before this was honest: variance must include observation noise; the kriging mean must actually be applied, since it is the correction the variance describes; and the prediction must be scaled by a factor fitted on one city and validated on another. See §E6.

C14. Survey planning by submodular maximisation

Given budget K , which locations minimise uncertainty across the *whole* city? With S the chosen set,

$$F(S) = \sum_i [\text{var}(x_i | \emptyset) - \text{var}(x_i | S)]$$

F is monotone and submodular — each observation helps, but helps less once nearby ground is covered — so greedy selection is within $(1 - 1/e)$ of optimal and exact solution is NP-hard [8].

The marginal gain has a closed form, which is what makes this fast:

$$\text{var}(\mathbf{x} | S \cup c) = \text{var}(\mathbf{x} | S) - \text{cov}(\mathbf{x}, c | S)^2 / (\text{var}(c | S) + \sigma_n^2)$$

$$\text{gain}(c) = \sum_i \text{cov}(x_i, c | S)^2 / (\text{var}(c | S) + \sigma_n^2)$$

so every candidate is evaluated against every parcel in one vectorised pass, with no iteration over candidates.

Candidates are parcel corners. Selection stops when marginal gain falls below 0.2% of the first point's — the system decides *how much* survey is worth buying, not just where.

A verification assertion checks that marginal gains are **non-increasing**. That is submodularity appearing as a testable property of the implementation rather than as a citation in a slide.

C14.1 Output shaped as a work order

Rank	Parcel	Lon	Lat	RMSE after	Δ	Reason
1	L01493	78.77917	28.45462	5.8271 m	-1.7043	highest-uncertainty region; anchors the whole sheet
2	L00647	78.77650	28.45397	5.0354 m	-0.7917	fills a gap between established control points
3	L02174	78.78103	28.45300	4.2905 m	-0.7449	fills a gap between established control points
4	L02430	78.78094	28.45659	3.5849 m	-0.7056	fills a gap between established control points
5	L00764	78.77639	28.45668	2.9061 m	-0.6788	fills a gap between established control points
6	L00843	78.77784	28.45188	2.5366 m	-0.3695	marginal refinement
7	L02840	78.78261	28.45496	2.2812 m	-0.2554	marginal refinement
8	L00061	78.77531	28.45237	2.1192 m	-0.162	marginal refinement

C15. Change detection

Epochs are compared **by spatial overlap, not by shared identifier**, because the identifier is exactly what cannot be trusted across a resurvey. DSM height delta separates a building that was *extended* (footprint grew) from one that was *heightened* (footprint stable, height rose) — a distinction that matters for property tax and for unauthorised construction, and that a 2-D pipeline cannot make at all.

Class	Signal	Count
New	footprint present in t1, absent in t0	31

Demolished	converse	28
Extended	footprint area grew beyond tolerance	70
Heightened	DSM delta with stable footprint	59
Encroachment	structure intersecting government land	9, 251.8 m ²

Output is a **change dossier, explicitly not a mutation record**. A mutation is a statutory act requiring notice and hearing whose form varies by state; generating one automatically would be both legally wrong and outside the problem statement.

C16. Audit chain

Each entry stores SHA-256 over canonical JSON of its payload plus the previous entry's digest:

$$h_i = \text{SHA256}(\text{canonical}(\text{payload}_i) \parallel h_{i-1})$$

`verify()` distinguishes a **content** break — payload no longer hashes to its recorded digest — from a **link** break, where an entry was removed or reordered. The two mean different things to an investigator, and reporting "chain invalid" for both is strictly less useful.

Computed **server-side**. A chain computed in the browser proves nothing; anyone can edit it in devtools. `/api/audit/tamper-demo` corrupts an entry deliberately so the failure can be shown rather than described.

C17. Access control and export

Role	Owner	Tenure	May act
Public	no	no	—
Surveyor	no	yes	raise for survey
Revenue clerk	yes	yes	escalate, raise for survey
Tehsildar	yes	yes	accept, reject, escalate, survey

Redaction is enforced **server-side**; owner names are personal data under the DPDP Act 2023 and a public caller never receives them. Withheld fields are *named* in a `_redacted` marker rather than silently dropped, so a consumer can distinguish withheld from missing. Verified live on the deployed instance: a public request returns `_redacted: [KHATEDAR_NM, AREA_BIGHA, TENURE_TYP]` where a tehsildar request returns the values.

Export produces GeoJSON, Shapefile with generated `.prj`, CSV and a PDF audit report under a SHA-256 manifest. DBF field names truncate at 10 characters, so the mapping is recorded in a sidecar rather than letting `KHATEDAR_NM` and `KHATEDAR_NO` silently collide. The OGC API — Features endpoint applies the same redaction: a standards-compliant interface is not a way around the access control.

Part D — Evaluation

How we know, including what we got wrong first.

D1. The ground-truth problem

Both [2] and [3] are limited by the same thing: no correct geometry exists to score against. [2] reports relative improvement across rounds; [3] reports a proxy (polygon-split counts). **Neither can state its error in metres.**

We generate a clean synthetic cadastre, retain it as the key, damage a copy through twelve recorded stages, and score recovery against the key. This follows the known-deformation construction of [2] exactly; the difference is that our deformations are drawn from the *documented* taxonomy of §A6.2 rather than chosen for convenience.

It also answers the review's "lack of large-scale, high-quality annotated data" [1] in the only way available: by constructing it.

D2. The damage model

#	Stage	Parameter	Real-world cause	In [5]
1	Similarity misregistration	shift (6.5, -4.2) m, rotation 0.85°, scale 1.0016	sheet georeferenced from too few, poorly spread GCPs	yes
2	Smooth non-linear warp	amplitude 2.4 m over 9 control points	paper shrinkage, scanner distortion, map-sheet join	yes
3	Vertex jitter	σ 0.35 m per vertex	manual digitisation from a paper sheet	yes
4	Vertex decimation	p = 0.10 per vertex	generalisation during digitisation	—
5	Sliver injection	6% of parcels shrunk 0.45 m	adjoining sheets digitised independently, leaving voids	yes
6	Overlap injection	5% of parcels grown 0.55 m	shared boundary double-counted between sheets	yes
7	Self-intersection	1.2% bow-tied; 4% duplicate vertices	careless digitising, unclosed rings	—
8	Schema drift	all 7 columns renamed and reordered	every department names its columns differently	—
9	Owner corruption	14% transliteration typos, 5% missing	variance across romanisations	—
10	Area unit drift	m ² re-expressed as bigha ($\div 2529.285$)	records kept in bigha and biswa	yes
11	Khasra renumbering	45% of blocks renumbered	resurvey reissues parcel numbers	—
12	Mutation	4% split, 3% merged, 2.5% deleted, 2% spurious	subdivision, amalgamation, record loss	—

Parameters quoted from the dataclass defaults in `synth/corruption.py`, so the table and the harness cannot disagree. "In [5]" marks the five stages that correspond directly to a failure mode documented by Sengupta et al.

D3. Experimental design

- **Two cities.** Model trained on one, all headline figures measured on a second with different layout and a different damage draw.
- **Group splits by source parcel**, never by pair (§C7), so a parcel's candidates cannot straddle train and test.
- **The targeting scale factor** is fitted on the train city and validated on the held-out one (§D7).
- **Fixed seed**, so every figure in this document is reproducible by running `scripts/build_demo.py`.

D4. Two experiments that changed the design

D4.1 Pipeline order

Matching before georeferencing gave blocking recall **0.58**. 42% of true correspondences never reached the model, and the model compensated by learning attribute joins. After reordering: **1.0**. The architecture of §C1 is the result of that measurement, not of intuition.

D4.2 The leakage trap

The first matcher scored **F1 0.9955** — better than our current figure — with khasra and owner supplying **96%** of the decision and IoU **0.24%**. It had learned to join on an identifier, which fails in any jurisdiction that renumbers on

resurvey: that is, in the exact situation the problem statement describes.

The harness now rennumbers 45% of blocks and genuinely changes 12% of owners, so the shortcut does not exist.

A better-looking number was replaced by a worse-looking number that means something.

D5. The verification suite

Eight stages, ~25 assertions, end to end (`scripts/verify_all.py`, 336 lines). Those that guard a claim this document makes:

Assertion	Guards against
<code>round-trip < 1 mm</code>	CRS drift
<code>central meridian == 500000 m</code>	UTM false-easting error
<code>agrees with pyproj < 1 mm</code>	our hand-rolled CRS being subtly wrong
<code>no overlaps in ground truth</code>	a dirty key, which would void every later figure
<code>damage actually applied</code>	scoring recovery on an undamaged copy
<code>coarse alignment reduces error > 40%</code>	georeferencing silently regressing
<code>blocking recall ≥ 0.98</code>	the 0.58 failure recurring unnoticed
<code>F1 ≥ 0.90</code>	matcher regression
<code>ECE ≤ 0.02</code>	confidence becoming decorative
geometry ≥ 70% of gain	the leakage trap returning
<code>exact set match ≥ 0.85</code>	assignment degrading on splits and merges
<code>spurious parcels rejected ≥ 0.95</code>	accepting records that should not exist
<code>topology errors reduced ≥ 90%</code>	cleaner regression
refuses to fill parcel-sized holes	the restraint of §C10.1 being optimised away
<code>lengthscale in 5-500 m</code>	a GP that fits but means nothing
<code>marginal gains non-increasing</code>	submodularity violated by an implementation bug
<code>optimised placement beats random</code>	the central targeting claim

Two of these exist because the property they protect is exactly what a well-meaning optimisation would remove. A model allowed to use khasra numbers scores *better*. A cleaner that fills every hole reports *fewer* errors. Both would be worse systems, and a test is the only thing standing between the project and that "improvement".

D6. Results

Full pipeline 45.6 s on one core; 3,047 legacy against 2,952 reference over 52,468 candidate pairs.

Stage	Measure	Result
Georeferencing	positional RMSE	8.4529 → 3.127 m
Schema	columns mapped	7/7
Schema	area unit from geometry	bigga @ 0.9672
Blocking	true pairs retained	1.0
Matching	precision / recall / F1	0.999 / 0.9997 / 0.9993
Matching	ROC AUC	1.0
Matching	TP / FP / FN / TN	2,996 / 3 / 1 / 49,468
Matching	decision from geometry	95.6%

Confidence	ECE / Brier	0.00009 / 0.00008
Assignment	1:1 / split / merge	2,791 / 148 / 5
Topology	total errors	6,168 → 215
Topology	overlapping pairs	5,038 → 0
Topology	doubly-claimed land	18,038.6 → 0.0 m ²
Restraint	parcel-sized gaps refused	69 of 73
Change	new / demol. / ext. / height.	31 / 28 / 70 / 59
Encroachment	sites on public land	9,251.8 m ²
Targeting	predicted / achieved / random	1.4479 / 1.3177 / 2.0567 m

D6.1 Where the decision comes from

Feature	Gain	Share
src_frac_in_tgt	0.46506	46.5%
centroid_dist	0.39963	40.0%
iou	0.06393	6.4%
kh_present	0.0174	1.7%
sym_diff_norm	0.01576	1.6%
kh_child	0.01027	1.0%
is_best_iou	0.00436	0.4%
vertex_ratio	0.00265	0.3%
iou_rank	0.00228	0.2%
d_compactness	0.00223	0.2%
centroid_dist_norm	0.00219	0.2%
rev_iou_rank	0.00179	0.2%

The three leading features — containment, centroid distance and IoU — are purely geometric and account for **92.9%** of total gain. `kh_present` ranks fourth at 0.0174, and note what it encodes: not *which* khasra number, but **whether one exists at all**. A missingness signal, not a join key.

D6.2 Reliability

Bin	n	Mean predicted	Observed
0.0–0.1	49,448	0.0	0.0
0.1–0.2	3	0.1364	0.0
0.2–0.3	17	0.2	0.0588
0.3–0.4	1	0.3201	0.0
0.8–0.9	1	0.8742	0.0
0.9–1.0	2,998	0.9991	0.9993

In the top bin, 2,998 pairs at mean predicted 0.9991 against observed 0.9993. That agreement, not the F1, is what licenses the word "confidence".

D6.3 The area conservation ledger

Summed parcel area falls **2.879%** across topology repair, which looks alarming until decomposed:

Quantity	Before	After	Reading
----------	--------	-------	---------

Summed parcel area	423,179.1	410,996.0	falls
Doubly-claimed land	18,038.6	0.0	eliminated
Union footprint	ground counted once		rises

The sum falls because land recorded as owned by two parties simultaneously is now counted once. **No land was created or destroyed, and the ledger demonstrates it rather than asserting it.** A system reporting only the total would appear to have lost 12,183 m² of someone's property.

D7. Targeting validation

K	Train predicted	Train achieved	Held-out predicted	Held-out achieved
10	1.2052	1.7853	2.0256	1.7508
20	0.9536	1.5344	1.6567	1.4535
40	0.8623	1.293	1.522	1.3485
80	0.8121	1.2261	1.4483	1.2715

Prediction scale **1.5046** fitted on the train city; held-out ratio median **0.8776** — the prediction errs *conservative* on unseen data, which is the direction an operational planning tool should err in. Irreducible floor **0.9009 m**: per-parcel digitising noise no quantity of control removes, reported rather than promised away.

Deployed result: 19 GCPs, baseline 3.127 m, predicted 1.4479 m, **achieved 1.3177 m**, random placement 2.0567 m — **35.9% better than random.**

Part E — System, deployment, engineering record

What was built, what it cost, and what the platform did to us.

E1. Composition

Component	Path	Lines	State
Harmonization engine	backend/ kshetra/	5,181	Verified by harness
API with RBAC	api/main.py	385	24 routes, deployed
Operator console	frontend/src/	3,698	Deployed
Build / verify / analysis	scripts/	4,433	Working

E1.1 Runtime profile

Stage	Seconds
generate city	0.69
inject damage	0.3
coarse georeference	0.07
blocking	4.02
features	16.2
train matcher	0.6
assign	0.15

topology repair	2.39
fit GP	16.44
plan survey	1.76
validate plan (optimised)	0.03
validate plan (random)	0.03
schema match	0.04
build epoch t1	0.01
change detection	0.1
encroachment	0.03

Feature construction and the GP fit dominate. Both are embarrassingly parallel and neither has been optimised; the figure is honest rather than tuned.

E2. Operator console

MapLibre GL JS v4, three Esri basemaps, `fill-extrusion` for surveyed building heights, and an alignment animation interpolating between raw and georeferenced layers so the correction is *watched*, not read about. Role tables in `store.ts` mirror `VISIBLE` and `MAY_ACT` in the API deliberately: the UI must not offer an action the server would refuse nor display a field it would withhold. The server remains the enforcement point; the client copy exists so the interface does not lie.

Sign-in reports the same error for an unknown username and a wrong password, because saying which half failed tells an attacker which usernames exist.

E3. Deployment

One Docker image serving the built UI and the API from a single origin. Two services would be simpler to reason about, but then page and API sit on different origins and the redaction and audit chain need CORS to agree with them.

Stage	Base	Does
1	<code>node:20-alpine</code>	<code>npm install</code> ; copy artifacts into <code>public/data</code> ; <code>tsc -b && vite build</code>
2	<code>python:3.12-slim</code>	install FastAPI/uvicorn/pydantic; copy backend, api, data, dist; run uvicorn

The API imports only `kshetra.audit` — `stdlib` `hashlib` and `json` — so nothing in the request path needs `numpy`, `scipy`, `shapely` or `xgboost`. Image drops from ~900 MB to ~200 MB.

Hosting-tier limits, not design properties: free instances spin down after 15 min idle with ~50 s cold start, and the audit chain writes to ephemeral disk, persisting within a session and resetting on redeploy.

E4. The platform problem

Windows **Smart App Control** blocked the scientific Python stack on the development machine — `pyproj`, `scikit-learn` and `LightGBM` at every version, and `numpy` 2.0.2 while permitting 2.4.6. Blocking is per-binary and undocumented in advance, presenting as an unexplained `ImportError` on a package that installed cleanly. On 26 September it was auto-promoted from evaluation to enforced and stopped everything at once.

Two consequences shaped the codebase, and both turned out to be improvements: the CRS engine was written from first principles and is now cross-validated to sub-millimetre (§C2), and **every optional dependency degrades gracefully** — metrics are hand-implemented, the `pyproj` check skips rather than fails. That is why the project survived the promotion.

Resolution was WSL2, where without `sudo` `ensurepip` is unavailable, so environments are built with `venv --`

without-pip plus get-pip.py, and Node from a tarball into ~/.local. SAC also blocks Git Bash's bundled ssh, so the repository points core.sshCommand at Windows OpenSSH.

E5. Defect register

Defect	What was wrong	Resolution
The leakage trap	F1 0.9955 with khasra and owner supplying 96% of the decision and IoU 0.24%. The model had learned to join on an ID, which fails wherever resurvey renumpers.	Harness renumpers 45% of blocks and changes 12% of owners. A test asserts geometry $\geq 70\%$ of gain. It is now 95.6%.
Blocking before georeferencing	Candidate generation on raw coordinates gave blocking recall 0.58; 42% of true pairs never reached the model.	Pipeline reordered. Recall 1.0.
GP over-optimism	Predicted 0.61 m against 1.23 m achieved. Variance quoted without observation noise; no kriging-mean correction; uncalibrated scale.	All three fixed and the scale validated on a held-out city. Reported honestly: better-than-random fell 51.8% $\rightarrow$ 35.9%, because a better interpolator also rescues random placement.
fitBounds over 400 features	Features are block-ordered, so the first 400 span a 170 m strip and the opening camera framed a sliver.	Bounds over the full collection.
Layer-visibility race	boot() added layers after the sync effect last ran; the uncertainty field rendered as a haze over everything.	ready flag added to the effect dependencies.
Percentage width on inline elements	Importance bars were spans; percentage widths are ignored on inline elements, so every bar rendered at 0 px.	Track and fill declared display:block.
GeometryCollection from gap fill	Occasionally produced a GeometryCollection, crashing the shapefile writer.	_largest_polygon() at the export boundary.
Sign-in overlay inside the stage	The toolbar showed a role before anyone had signed in.	Overlay moved to position:fixed with its own stacking context.
Basemap maxzoom	Esri canvas basemaps stop at z16; switching while pitched at z16.8 requested nonexistent tiles and blanked the map.	Per-basemap maxzoom declared.
Command-palette PII leak	Owner names filtered from palette search for the public role but still rendered as result subtitles. Found by using the system as public, not by reading it.	redact() moved to the point of render.
autocrlf rewrote shell scripts	Line-ending conversion broke every path in .sync.sh under bash.	.gitattributes pins LF for .sh, .py, Dockerfile, YAML.
fpdf2 core fonts are latin-1	The audit report rejected an em-dash and crashed the export.	_PDF_FOLD maps out-of-range characters to ASCII.

E5.1 The one worth dwelling on

The GP over-optimism fix is the defect we are most inclined to report, because correcting it made a headline number **worse**. Better-than-random fell from 51.8% to 35.9% — not because the planner degraded, but because a better interpolator also rescues random placement, so the old margin was partly an artifact of the bug. **A system whose reported advantage falls when its model is corrected is a system whose numbers were previously flattering it.**

Part F — Assessment

Where this stands, and where it does not.

F1. Comparison against prior art

Capability	[2] Girard	[3] Suwardhi	FME/Esri	KSHETRA
Corrects misregistration without ground truth	yes	yes	no	yes
Reports error in metres against a key	no	no (proxy)	n/a	yes
Multi-source (n > 2)	no	no	yes	yes
Infers attribute schema	no	no	no (human declares)	yes
Infers area unit from geometry	no	no	no	yes
Calibrated per-parcel confidence	no	no	no	yes
Separates measured from recorded authority	no	no	no	yes
Detects split / merge explicitly	no	partial	no	yes
Conserves and proves area	no	no	no	yes
Declines to fill parcel-sized gaps	n/a	n/a	no	yes
Plans where to survey next	no	no	no	yes
Validates that plan against random	no	no	no	yes
Hash-chained audit	no	no	no	yes
Evaluated on real cadastral data	yes	yes	yes	no

The final row is the one that matters against us, and it is placed last deliberately rather than omitted. Girard and Suwardhi evaluate on real cadastral layers; we evaluate on synthetic ones because no bulk Indian urban cadastre is public. Our advantage in measurability is purchased exactly by that limitation.

F2. Limitations

- The cadastral parcel layer is **synthetic**, derived from OpenStreetMap block structure and labelled as such in the product and its exports. Building footprints, road network and terrain are real.
- Encroachment cases are **deliberately injected**; the generator never places structures on public land, so without injection the claim would be untested.
- The survey-plan prediction is **calibrated, not exact** — ~12% transfer error to a held-out city, erring conservative.
- Per-parcel digitising noise is **irreducible** by survey control; the system reports the floor rather than promising accuracy below it.
- The GP kernel is **smooth** and cannot represent a sheet-join discontinuity — the same limitation [2] states for its displacement model.
- The system produces a **change dossier, not a mutation record**.
- Redaction is enforced server-side; in the static demonstration build every artifact reaches the browser, so redaction *there* is a display control only.
- Evaluation is on two synthetic cities. **Generalisation to a real ward is untested** and we do not claim it from these numbers.
- The Docker build stages have been reproduced by hand but the container has not been run in CI.

F3. Threats to validity

Threat	Why it is a threat	What we did
Construct — the damage model might not resemble real damage	If so, recovery figures say nothing about real sheets	Five of twelve stages map directly onto Sengupta's documented failure modes; parameters are in the open in §D2 for a reviewer to dispute
Internal — leakage between train and test	Would inflate every matching figure	Group splits by source parcel; the geometry-share assertion; the leakage trap of §D4.2 was caught this way
External — synthetic to real transfer	Real cadastres have structure our generator does not model	Not mitigated. Stated as the principal limitation
Conclusion — one damage draw, one seed	Figures could be a lucky sample	Held-out city with an independent draw; targeting validated across four budgets. No repeated-seed variance study — a real gap

F4. Roadmap

1. **Repeated-seed variance study** — the cheapest material improvement to the evidence base, and named as a gap above.
2. **Sliver classification with legal consequence** — area, compactness, khasra presence, RoR presence, footprint presence. Currently geometry alone.
3. **Per-state policy objects** — FMB-primary regimes must not have geometry rewritten as authority (§A2).
4. **Transformation-order selection benchmarked against [5]'s 310 sheets** — would convert §C3 from a design claim to a measured one.
5. **A released Indian benchmark** for one pilot ULB.

F5. Open questions we would not claim to have solved

- **Invisible boundaries.** Fusing imagery, legacy geometry, textual record and utility topology to infer a boundary with no physical trace. The ceiling of the vision-only approach, and unaddressed in the literature we found.
- **Discontinuous displacement fields.** Both we and [2] assume smoothness; sheet joins are discontinuities by construction.
- **Legal admissibility.** Nothing here establishes that a calibrated confidence is evidence in a revenue court. That is a question for jurists, and we flag it rather than assuming the answer.

Appendix A — module inventory

Tracked first-party Python over 90 lines.

File	Lines
backend/kshetra/synth/corruption.py	464
api/main.py	385
backend/kshetra/topology/cleaner.py	384
backend/kshetra/attributes/schema_match.py	348
backend/kshetra/export/writers.py	325

backend/kshetra/synth/generator.py	306
backend/kshetra/change/detect.py	299
backend/kshetra/conflict/resolve.py	299
backend/kshetra/targeting/uncertainty.py	278
backend/kshetra/targeting/planner.py	275
backend/kshetra/georef/transform.py	254
backend/kshetra/matching/assign.py	250
backend/kshetra/crs.py	231
backend/kshetra/matching/features.py	230
backend/kshetra/matching/model.py	200
backend/kshetra/attributes/text.py	191
backend/kshetra/georef/coarse.py	186
backend/kshetra/audit.py	174
backend/kshetra/geometry.py	170
backend/kshetra/layers.py	154
backend/kshetra/evaluation/metrics.py	148

Appendix B — glossary

Term	Meaning
khasra	Parcel number in north Indian revenue records. Carries a recursive sub-division grammar on subdivision.
khatauni / jamabandi	Record of rights. Khatauni in UP; jamabandi in Punjab and Haryana.
patta / chitta	Tamil Nadu record of rights and its extract.
khata	Ambiguous and hazardous. A revenue holding account in the north; a municipal property-tax account in urban Karnataka.
khatedar	The recorded holder of a parcel.
mouza	Revenue village; the unit a cadastral sheet typically covers.
bigha / biswa	Traditional area units. Non-standard across states; bigha here is 2529.285 m ² .
karam	Linear unit underlying some area measures. Digitised in centimetres in Himachal, which is why Bhu-Naksha carries a ×22 scale factor there.
FMB	Field Measurement Book. The legally primary record in southern states, where the map is a derived sketch.
RoR	Record of Rights — the textual register of who holds what.
mutation	The statutory act of updating the record after a transfer. Requires notice and hearing; form varies by state.
tehsildar	Revenue officer with authority to resolve a contested parcel.
ULPIN	Unique Land Parcel Identification Number — 14 characters generated from geo-coordinates under DILRMP.
CORS	Continuously Operating Reference Station. NAKSHA specifies control better than 5 cm.
GSD	Ground Sample Distance — imagery resolution. NAKSHA specifies 5 cm.

DSM / DTM	Digital Surface / Terrain Model. The difference gives building height, which separates an extended building from a heightened one.
LADM	Land Administration Domain Model, ISO 19152. A formal language for describing land administration, not a data product spec.

Appendix C — commit history

Commit	Date	Subject
385a199	2026-10-03	Full project dossier -- 22 pages, every figure read from the artifacts
fc84d56	2026-09-30	Pin LF on Dockerfile and YAML
b67e296	2026-09-30	Deploy: single Docker image serving the UI and the API on one origin
67af6d5	2026-09-30	Expand the research summary from 4 pages to 9
c9cb84a	2026-09-30	Research summary PDF with paper-backed positioning
0c252f1	2026-09-30	Credential sign-in; 3D map, basemap switcher, bolder cartography
764682e	2026-09-29	Slide 1 and 3 mockups in the team's existing SIH idiom
4e013cc	2026-09-29	Role sign-in, KSHETRA mark, command palette
f38ca3a	2026-09-29	Round the figures in the demo script
7999a34	2026-09-28	Guided demo mode; no-cache server
0027ad7	2026-09-28	Light theme; basemap degrades gracefully; error boundaries
8d30992	2026-09-28	Calibrate the survey-plan prediction; correct with the GP, not a spline
394f0e3	2026-09-28	Add Export dock tab and a standalone launcher
fafba65	2026-09-28	Tier 2: server-side audit chain, export writers, FastAPI with RBAC
7d76d4c	2026-09-28	Pin LF endings for shell scripts; drop one-shot patch scripts
895c375	2026-09-28	Rename to KSHETRA; add schema matching, conflict resolution, change detection
252a9a9	2026-09-28	Working MVP: engine verified, demo artifacts, mission-control UI
d8e7fd7	2026-09-27	Add PROMPT.md -- single pasteable build prompt
5d10eb0	2026-09-27	Deepen build prompt: conflict taxonomy, interaction spec, failure modes
f719fd7	2026-09-27	Add styled handoff page with print stylesheet
ca532b7	2026-09-27	Add HANDOFF.md for device migration
2a53916	2026-09-27	BhoomiSetu: engine, targeting module, real data, build plan

Appendix D — reproduction

```
python scripts/build_demo.py    # regenerates every artifact
python scripts/verify_all.py    # ~25 assertions, 8 stages
python scripts/calibrate_targeting.py
python scripts/build_compendium.py # this document
```

Fixed seed throughout. Engine requires numpy, scipy, shapely and xgboost; the serving path requires none of them.

Appendix E — references

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- [5] A. Sengupta, C. Lemmen, W. Devos, D. Bandyopadhyay and A. van der Veen. *Constructing a seamless digital cadastral database using colonial cadastral maps and VHR imagery — an Indian perspective*. Survey Review, 48(349):258–268, 2016. doi:10.1179/1752270615Y.0000000003.
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- [13] Department of Land Resources. DILRMP progress statistics.
- [14] DAKSH. *Access to Justice Survey 2015-16*. 9,329 litigants, 305 locations, 24 states.
- [15] Tamil Nadu Department of Survey and Settlement. DILRMP Regional Review, Bengaluru, 6 September 2024.
- [16] ISO 19152-1:2024, Land Administration Domain Model — Part 1: Generic conceptual model.
- [17] A. Sengupta, D. Bandyopadhyay, C. H. J. Lemmen and A. van der Veen. *Potential use of LADM in cadastral data management in India*. 5th LADM Workshop, Kuala Lumpur, September 2013.

Appendix F — claims we deliberately do not make

- **"Land disputes take about 20 years to resolve."** Attributed to NITI Aayog across many secondary sources; original not opened. [S]
- **"324 years to clear the case backlog."** Source not located. [U]
- **"25% of Supreme Court decided cases involve land disputes."** Widely repeated, original study not found. [U]
- **Any figure for man-hours or rupee cost of manual GIS harmonisation** specifically, as distinct from survey or digitisation cost. We searched and found nothing credible and do not invent one; where a proxy is needed we use the official ₹56,725/sq km resurvey rate [15].
- **"Indian Geodetic Datum 2023."** Does not exist. NGP 2022 [7] commits to *redefining* the framework; the Indian Vertical Datum is real and referenced in [6].
- **The exact 14-character composition of ULPIN**, and the contents of the NAKSHA SDMS schema, whose SOP is available only as an untexted scan. [U]

Sources [1]–[4] and [8] were read in the original. [5]–[7] and [9]–[17] are drawn from a sourced research dossier compiled for this project; figures carry their grade, and any figure intended for publication should be re-checked against the primary document.